Q3 2024 CS 497 Capstone Project

Enhancing Fraud Detection Through Hybrid Modeling

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ABSTRACT

With the global rise in digital transactions, the need for robust fraud detection has never been more urgent as this global reach is available to remote actors. Traditionally, transactional fraud would have a physical aspect of it being something like a chargeback or product scam, skimmer, or overpayment-charge card scam. These scams could only be assessed through explicit transactional data including transaction amount, merchant IDs, and timestamps after the event (Lee, Part I, 2017). To protect customer privacy and reduce exposure risks, many datasets are anonymized and assessed using transformed features through techniques like Principal Component Analysis (PCA). I will utilize past research and leverage a dataset provided by Machine Learning Group via Kaggle containing such transactional data anonymized under PCA to find underlying patterns and trends while preserving privacy via hybrid modeling techniques advised by previous researchers. The supervised modeling techniques covered in this study include logistic regression, support vector machines, random forests, neural networks, and hybrid/ensemble techniques to enhance fraud detection.

Keywords: PCA, Hybrid modeling, Kaggle, Neural networks, support vector machines, random forests

INTRODUCTION

Credit card fraud is a growing concern in today's cashless society. Machine learning has become a promising tool in modernized fraud detection addressing more complex and/or subtle behavior. In past studies, these patterns have been identified through supervised techniques in machine learning like logistic regression, support vector machines, random forest classifiers, and deeper learning models like artificial neural networks (FNN, CNN, GNN). Following approaches taken by related studies that demonstrate these models' potential for high precision and recall, researchers have begun experimenting with hybrid modeling, blending various algorithms to enhance detection rates. My focus will be to share and evaluate the findings of previous authors and implement their results across multiple models to improve this solution in credit card fraud detection.

Problem Statement

The objective is to develop an effective fraud detection model using anonymized features from transaction data. By leveraging traditional machine learning techniques, the goal is to accurately identify fraudulent activities without relying on explicit interpretation.

Motivation

Credit card fraud is a growing concern in today's cashless society. As Digital security expert Brett Cruz emphasizes in his study, an estimated 52 million Americans are affected by fraudulent charges totaling up to USD 5 billion (Cruz, 2024). As spending increases, so do the chances and impacts of fraudulent activity. To prevent an increase in the number of attacks and the amount of money stolen, we must highlight the need for better and faster detection.

Conclusions

Between the studies of linear and nonlinear techniques, I will combine some of the insights and types of learning to create hybrid, nonlinear approaches to detecting fraud in the datasets. The nonlinearity will be key in looking for multiple complex and subtle influences in classifying fraud.

This study aims to shed further insight into modeling behavior under credit card fraud detection. The goal is to fuse past studies to continue refining models toward perfection.

BACKGROUND

Data Preprocessing

Clean and structured data is crucial for optimal model performance. Without careful preprocessing outliers, inconsistencies, or missing values can skew predictions. Additionally, the attributes or features will be filtered to ensure the right context is fed for distinguishing between classes. When certain values range too far, they can be scaled to a smaller range to maintain interpretation but reduce the influence of outliers.

The data provided by Kaggle contains credit card transactions from an anonymous bank containing over 280,000 rows of data. As mentioned, the data is masked under PCA, a statistical technique used to reduce the dimensionality while preserving as much variance as possible (Jolliffe and Cadima, 2016). After reviewing the 31 features I have decided to remove ‘time’ as it doesn’t offer significance. The data set is heavily imbalanced with 99.83% consisting of non-fraudulent records. Furthermore, a heatmap will validate the use of models in this study as the data is highly uncorrelated:

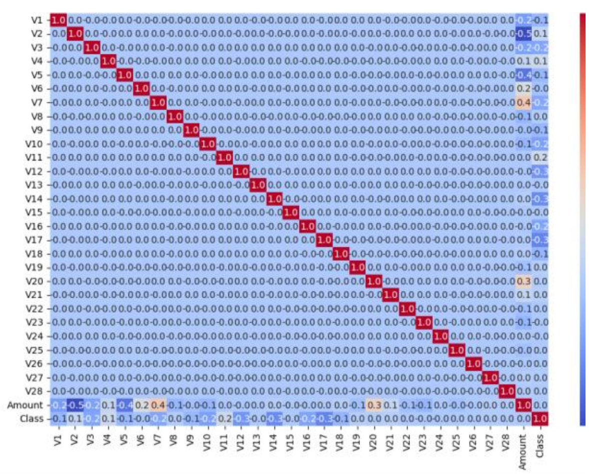

Figure 1 - Heatmap

Traditional Models

Support Vector Machine (SVM) is a technique where all training data remains available for the model to build support vectors or subsets of the data to define the margins of the hyperplane used in classification for separating classes. The complexity of such models is based on their number of support vectors rather than the dimensionality of the dataset. To handle nonlinear data, SVM utilizes kernels (linear, polynomial, Radial Basis Function (RBF)) transforming it into a higher-dimensional space making it linearly separable. Kernels calculate a similar measure for the vectors providing inner products without the need to explicitly map to higher dimensional space (Awad and Khanna, 2015). Given that fraud detection involves high-dimensional, anonymized data, SVM is suitable for this study. Its use of support vectors allows it to handle complex feature-rich sets more efficiently.

Another suitable traditional technique would be the use of Random Forest classifiers (RF). RFs have proven to be the preferred traditional technique amongst many researchers on fraud detection. RF builds multiple decision trees using bootstrapping (sampling and replacement) and aggregates the results. The use of bagging or bootstrap aggregation ensures that each tree is trained on a slightly different subset of the data, leading to more diverse trees. The generalization error will be critical to evaluating how well the model can handle new, unseen data.

Deep Learning

Unlike traditional learning models which require manual feature engineering, NNs can automatically learn complex representations in their hidden layers. A common Feedforward Neural Network (FNN) processes input features through multiple hidden layers, applying weights and optimization algorithms like Adam. To refine the results, activation functions like ReLU, Sigmoid, or Softmax are applied at each node. Rectified Linear Unit (ReLU) is effective as it introduces non-linearity by outputting the input directly if positive otherwise zero. While ReLU is useful in learning, Sigmoid will be useful as the output layer providing probabilities for binary classification.

Weights are a fundamental component of Neural Networks (NN). They determine how the input data influences the output, guiding the learning process by shaping the model’s predictions (Awad and Khanna, 2015). Each connection between the nodes in different layers has an associated weight. When an input feature passes through a layer, it is multiplied by its respective weight, determining its influence on the next layer.

Additionally, regularization methods like dropout layers and L2 regularization can prevent the network from learning too much from noisy data or irrelevant patterns. This happens by having randomly selected nodes disabled to reduce the reliability improving generalization (Ketkar and Moolavil, 2021).

Thresholds will fluctuate when drawing predictions to determine how strongly the model is classifying the probabilities balancing precision and recall and showcasing effectiveness for different scenarios.

To determine the effectiveness of a neural network, the Receiver Operating Characteristic Area Under the Curve (ROC-AUC) score will be used. The ROC-AUC score provides a single metric that sums up the model's ability to distinguish between classes by evaluating the trade-off between sensitivity (true positive rate and specificity (false positive rate).

Hybrid Modeling

Hybrid Modeling can enhance the predictive nature of models as it is an integration of multiple modeling techniques. As credit card fraud continues to evolve to find new ways for success so should the model detecting it. Using variants of NNs the data fed can appropriately be everchanging providing new patterns for an NN's adaptability to change. Support Vector Machines and Random forests can obtain significant features in imbalanced datasets. These identified features can then be the input for an NN to classify fraudulent and legitimate transactions. A common way of achieving hybrid models is through a core idea in ensemble learning, stacking. In this type of learning, individual models or learners produce predictions which are then fed to a meta-model for final predictions. There are other ensemble techniques like bagging, blending, or boosting, but variations of stacking should suffice for this study. After assessing the results, I will utilize metrics in the minority class to evaluate model performance including cross-validation when on final models.

RELATED WORK

Previously, researchers have leveraged various data sets including data anonymized under PCA. The studies conducted used traditional techniques such as logistic regression (LR), Support Vector Machine (SVM), random forest classifiers (RF), and deep learning techniques like Neural Networks (NN) to achieve promising metrics producing a few trends. Due to the imbalanced nature of multiple datasets obtained, techniques such as Synthetic Majority Over-sampling TEchnique (SMOTE) or Adaptive Boosting (AdaBoost) have been applied to achieve more promising results (Cherif et al, 2024).

Simplified processes like binary classification in logistic regression have proven to perform the lowest in their F_score achieving around ~.50-.60 (Sharma et al, 2021). The SVM model's performance was generally better than logistic regression but scored lower in precision compared to due to the higher rate of FN. Evaluating RF models in Sharma's study, they resulted in an average F_score of .85, evidence of low FP and FN, while maintaining a higher precision of 82% (Sharma et al, 2021).

To deal with data imbalance in an extensive dataset, the research in a Journal of King Saud University states researchers applied under-sampling techniques to achieve a high accuracy of 96%, noting an area under the ROC curve of .989 (Cherif et al, 2024) with their RF model.

Transitioning into deeper learning and variations of neural networks. These models performed exceptionally well in capturing nonlinear interactions but in a different manner. While traditional techniques like Random Forests and SVMs are effective in capturing nonlinear interactions, deep ML models thrive in accounting for various features. Dealing with a dataset containing 28 features, Sharma et al. (2021) reported that their NN achieved an F1 score of .91, showcasing its ability to handle complex datasets.

Model	Accuracy	Precision	Recall	F_Score
Logistic Regression	0.94	0.54	0.92	0.57
Support Vector machine	0.95	0.76	0.80	0.78
Random forest	0.98	0.82	0.90	0.85
Neural Network	0.99	0.93	0.88	0.91

Figure 2.1 – Metrics from Pratyush Sharma et al (2021) study

Traditionally used for imaging data, the convolutional neural network (CNN) and hybrid CNN-SVM models by Berhane resulted in an F1_score of .82 and .90 technique (Berhane et al, 2023).

Proposed methods	Accuracy	Precision	Recall	*F1*-score	AUC
CNN	89.74	88.60	76.90	82.34	89.49
CNN-SVM	91.08	90.50	90.34	90.41	91.05

Figure 2.2 – Results from Berhane with hybrid modeling

The Graph Neural Network (GNN) referenced at King Saud University resulted in an F1 score of .86 (Cherif et al, 2024). The tradeoff of going into deeper learning for insights is that it requires more computational power and resources as opposed to something like RF and other traditional techniques. These key findings will facilitate the architecture for my hybrid models.

APPROACH

I will evaluate multiple models like SVM, RF, FFNN, and CNN using different techniques including a hybrid and ensemble approach. Dealing with class imbalance, I may employ SMOTE or under-sample the majority class. All models will feature dynamic learning rates, early stopping callbacks, and a range of batch sizes to optimize performance. Hyperparameter tuning will cover key elements such as kernel size, number of neurons/filters, and layers for diverse configurations. After evaluating the traditional models, I will create variations of neural networks and identify the best-performing architecture for combining models. Finally, I will experiment with ensemble methods to build a meta-model that combines the strengths of the individual models for enhanced classification accuracy.

DATA COLLECTION

After experimenting with various ratios of the class through SMOTE and/or under-sampling, I have chosen to only utilize one. Regardless of studies opting in SMOTE to sample the minority, much of the data in the non-fraud class still skews the learning process. Therefore, I will dynamically under-sample the majority while keeping the real fraud data unaltered.

SVM

```
svm_original = SVC(kernel='rbf', probability=True,
                   C=1.0, gamma='scale',
                   random_state=42, verbose=True)
```

Figure 3.1 – SVM Parameters

The 'rbf' kernel in SVM is to map the input data into a higher-dimensional space effectively handling non-linear classification problems. The kernel parameter 'C' is set to 1 allowing some degree of misclassification to prevent overfitting. The 'gamma' parameter controls the influence of a single data point. Scaling gamma allows more flexibility throughout the influence of data. By aggressively under-sampling the majority class, the SVM model produces the following results:

```
Performance on Resampled Dataset:
[[56850    14]
 [   44    54]]
              precision    recall  f1-score   support

           0       1.00      1.00      1.00     56864
           1       0.79      0.55      0.65        98

    accuracy                           1.00     56962
   macro avg       0.90      0.78      0.83     56962
weighted avg       1.00      1.00      1.00     56962

ROC-AUC Score: 0.97
```

Figure 3.2 – SVM Report

Although the performance here has improved compared to previous studies, it isn't significant enough to integrate with other models.

RF

```
rf_model = RandomForestClassifier(
    n_estimators=300,
    random_state=42,
    class_weight='balanced'
)
```

Figure 4.1 – RF model architecture

The RF model follows the same approach as standard architecture. Like the SVM, the RF model will use the probability of belonging to a class rather than the class label itself. The RF model was configured with more trees for voting power and training time at the expense of computational cost. The model has a balanced weight parameter that automatically adjusts the weights of classes according to their frequency in the dataset, targeting imbalanced data.

```
Confusion Matrix:
[[56858     6]
 [   19    79]]

Classification Report:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00     56864
           1       0.93      0.81      0.86        98

    accuracy                           1.00     56962
   macro avg       0.96      0.90      0.93     56962
weighted avg       1.00      1.00      1.00     56962

ROC-AUC Score: 0.9750
```

Figure 4.2 – RF Report

The metrics from the report align with the results of past studies showcasing the effectiveness of RF in detecting fraud. The rate of TN and TP is also much higher when compared to previous SVM models supporting the RF's strength in handling complex interactions.

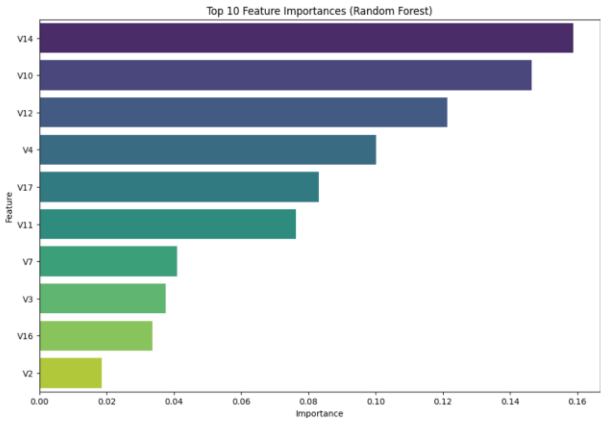

Figure 4.3 – Top 10 influencing factors in classification

The bar graph above shows us the top ten most influential factors in the classification decision providing insights into the data features that contribute the most to fraud detection.

NN

Transitioning into deeper learning, I started by creating a simple Feed Forward Neural Network (FFNN). This FFNN processes data sequentially through connecting layers making it a good fit for anonymized data. By experimenting with key hyperparameters such as the number of hidden layers or layers of neurons, optimizers, activation functions, and learning rates I was able to fine-tune the FFNN achieving in the following metrics:

```
Confusion Matrix:
[[56857     7]
 [   19    79]]

Classification Report:
              precision    recall  f1-score   support

     Class 0       1.00      1.00      1.00     56864
     Class 1       0.92      0.81      0.86        98

    accuracy                           1.00     56962
   macro avg       0.96      0.90      0.93     56962
weighted avg       1.00      1.00      1.00     56962

ROC-AUC Score: 0.9826223398757364
```

Figure 5.1 – FFNN Report (Threshold = .60)

The results are identical to the RF model in terms of overall performance with a difference only in computation. After achieving satisfactory results from the FFNN.

I transitioned to a Convolutional Neural Network (CNN). While CNNs are used for image data, they can be equally effective with structured data when aiming to capture local patterns and reduce dimensionality. Initially, the data had to be reshaped into 3D (samples, features, depth = 1), a low depth as sequential features were removed. The CNN introduces convolutional layers followed by pooling layers to focus on localized patterns, which could be beneficial with the multitude of features.

```
Confusion Matrix:
[[56851    13]
 [   20    78]]

Classification Report:
              precision    recall  f1-score   support

     Class 0       1.00      1.00      1.00     56864
     Class 1       0.86      0.80      0.83        98

    accuracy                           1.00     56962
   macro avg       0.93      0.90      0.91     56962
weighted avg       1.00      1.00      1.00     56962

ROC-AUC Score: 0.9740462026116017
```

Figure 5.2 – CNN Report (Threshold = .60)

The result of the CNN has proven solid performance but weaker than the FFNN, suggesting some tuning before additional modeling.

RF-CNN

Resembling Berhane's CNN-SVM, I directly combined the previously trained RF and CNN models to leverage the RF's feature extraction capabilities and the CNN's pattern recognition strengths. Utilizing the previously trained models I decided to feed the RF model to the CNN model leveraging RF's feature extraction and the CNN's pattern recognition process. Combining the two, I aim to obtain each model's complementary strength. After increasing the generation of trees, overall training by epochs, and experimenting with the CNN architecture, the model produced the following metrics at two different thresholds for the probability of being classified as fraud:

```
Confusion Matrix:
[[56852    12]
 [   17    81]]

Classification Report:
              precision    recall  f1-score   support

     Class 0       1.00      1.00      1.00     56864
     Class 1       0.87      0.83      0.85        98

    accuracy                           1.00     56962
   macro avg       0.94      0.91      0.92     56962
weighted avg       1.00      1.00      1.00     56962

ROC-AUC Score: 0.9714065353209376
```

Figure 6.1 – RF-CNN Model Report (Threshold = .60)

```
Confusion Matrix:
[[56857     7]
 [   19    79]]

Classification Report:
              precision    recall  f1-score   support

     Class 0       1.00      1.00      1.00     56864
     Class 1       0.92      0.81      0.86        98

    accuracy                           1.00     56962
   macro avg       0.96      0.90      0.93     56962
weighted avg       1.00      1.00      1.00     56962

ROC-AUC Score: 0.9714065353209376
```
Figure 6.2 – RF-CNN Model Report (Threshold = .73)

Although the hybrid model shows promising results, identical to the standalone RF model's metrics, the complementing strengths may not be leveraged appropriately.

RF-CNN-LR

Although previous studies showcased average results with LR, they perform well when applied to linearly separable problems. In this approach, I took the predictions of the RF model, and the CNN model alone served as a high-level representation of the data. The LR model was then used as the final decision-maker based on the ensemble features. Certain changes were made to the hyperparameters of the CNN model like increasing the filter size to look at half the features per step and increasing the weight of the minority class to 1:3.

```
Confusion Matrix:
[[56858     6]
 [   16    82]]

Classification Report:
              precision    recall  f1-score   support

     Class 0       1.00      1.00      1.00     56864
     Class 1       0.93      0.84      0.88        98

    accuracy                           1.00     56962
   macro avg       0.97      0.92      0.94     56962
weighted avg       1.00      1.00      1.00     56962

ROC-AUC Score: 0.9822
```
Figure 7 – RF-CNN-LR Meta model (Threshold = 0.14)

The overall performance is consistent with other models. Showing marginal improvements may suggest that the dataset's characteristics have fully utilized the capabilities of the standalone models.

RF-FFNN-LR

Lowering computational cost, I developed an RF-FNN-LR meta-model ensemble. The RF and FFNN proved their strong capabilities in distinguishing between classes, and LR will once again serve as the meta-classifier.

```
Confusion Matrix:
[[56860     4]
 [   18    80]]

Classification Report:
              precision    recall  f1-score   support

     Class 0       1.00      1.00      1.00     56864
     Class 1       0.95      0.82      0.88        98

    accuracy                           1.00     56962
   macro avg       0.98      0.91      0.94     56962
weighted avg       1.00      1.00      1.00     56962

ROC-AUC Score: 0.9903595618044628
```
Figure 8.1 – RF-FFNN-LR Meta model (Threshold = 0.30)

```
Confusion Matrix:
[[56862     2]
 [   20    78]]

Classification Report:
              precision    recall  f1-score   support

     Class 0       1.00      1.00      1.00     56864
     Class 1       0.97      0.80      0.88        98

    accuracy                           1.00     56962
   macro avg       0.99      0.90      0.94     56962
weighted avg       1.00      1.00      1.00     56962

ROC-AUC Score: 0.9899680081655622
```
Figure 8.2 – RF-FFNN-LR Meta model (Threshold = .85)

RESULTS

Data Analysis

The individual models and their hybridization have provided key insights into the trade-offs between precision, recall, and overall model performance with the minority class. Each stage of modeling highlights different strengths and limitations but there are several trends.

Support Vector Machine

MODEL	PRECISION	RECALL	F1-SCORE	CONFUSION MATRIX	ROC-AUC
SVM	0.96	0.47	0.63	[[56862, 2], [52, 46]]	0.96
SVM (Under-sampling)	0.79	0.55	0.65	[[56850, 14], [44, 54]]	0.97

Figure 9.1 – Both SVM Reports

- The SVM struggled with imbalanced data showing bias in the majority class, this is evident by the low recall of .55 in the minority class.
- This model performed significantly better with aggressive under-sampling compared to techniques like SMOTE aligning with a study conducted at King Saud University mentioning the risk of added noise and overfitting (Cherif et al, 2024).
- This model identifies real transactions but has unacceptable recall (detecting minority class).

Random Forest Classifier

MODEL	PRECISION	RECALL	F1-SCORE	CONFUSION MATRIX	ROC-AUC
RF	0.97	0.73	0.84	[[56862, 2], [26, 72]]	0.95
RF (Under-sampling)	0.93	0.81	0.86	[[56858, 6], [19, 79]]	0.97

Figure 9.2 – Both RF Classifier Reports

- Utilizing weights and under-sampling, the RF significantly improved in its generalization
- Additionally, the RF was able to provide feature importance highlighting V (14, 10, 12, 4, 17, 11) as the most influential in predictions. Businesses providing such data can reverse the anonymity of features to understand factors in fraud detection further.

Feed Forward Neural Network

MODEL	PRECISION	RECALL	F1-SCORE	CONFUSION MATRIX	ROC-AUC
FFNN	0.92	0.81	0.86	[[56857, 7], [19, 79]]	0.98

Figure 9.3 – FFNN (Threshold = .60)

- Showcased adaptive learning producing more than satisfactory metrics with a slight trade-off between precision and recall (.92 and .81)
- Despite being a solid model, the FFNN underperforms in comparison to the analysis done by P. Sharma (Sharma et al, 2021).

Convolutional Neural Network

MODEL	PRECISION	RECALL	F1-SCORE	CONFUSION MATRIX	ROC-AUC
CNN	0.86	0.80	0.83	[[56851, 13], [20, 78]]	0.97

Figure 9.4 – CNN (Threshold = .60)

- The precision was lower than the previous two models detecting more false positives in legitimate transactions.
- The increase in computation costs may not be effective in maximizing the potential of this data.

RF-CNN Hybrid

MODEL	PRECISION	RECALL	F1-SCORE	CONFUSION MATRIX	ROC-AUC
RF-CNN (.6)	0.84	0.83	0.84	[[56849, 15], [17, 81]]	0.98
RF-CNN (.76)	0.91	0.79	0.84	[[56856, 8], [21, 77]]	0.98

Figure 9.5 – RF-CNN

- The hybrid model at .60 threshold provided a balanced macro average of .92 and .91 effectively discriminating between classes and capturing many fraudulent transactions.
- The model shows marginal improvements to past models as they might need further methods of meta-modeling to optimize the use of this dataset.

RF-FFNN-LR Meta Model

MODEL	PRECISION	RECALL	F1-SCORE	CONFUSION MATRIX	ROC-AUC
RF-FFNN-LR (.3)	0.95	0.82	0.88	[[56860, 4], [18, 80]]	0.99
RF-FFNN-LR (.85)	0.97	0.80	0.88	[[56862, 2], [20, 78]]	0.99

Figure 9.6 – RF-FFNN-LR Meta Model

- The meta-model performs hyper consistently across threshold indicating it may not be as adaptable for multiple scenarios.
- The model showcases the strongest discriminative power with an ROC-AUC score of .99 and consistently high F1-scores highlighting its strong precision in recognizing TN very well.
- The model may be a better solution when compared to the previous meta-model as it utilizes fewer resources in complexity leading to reduced training times. The model also correctly identifies more samples.

RF-CNN-LR Meta Model

MODEL	PRECISION	RECALL	F1-SCORE	CONFUSION MATRIX	ROC-AUC
RF-CNN-LR (.14)	0.93	0.84	0.88	[[56858, 6], [16, 82]]	0.98
RF-CNN-LR (.46)	0.95	0.83	0.89	[[56860, 4], [17, 81]]	0.99

Figure 9.7 – RF-CNN-LR Meta Model

- Off initial evaluation, all metrics have marginal but steady improvements. The model outperforms all other models, even reducing the total number of FP and FN with a strong ROC-AUC score of .98/.99.
- Prioritizing recall, the model misclassified a smaller 16 transactions as non-fraud while misclassifying only 6 as fraud.
- At a balanced threshold near default, the model has produced the strongest metrics

Overall, the LR meta-models are the best performers. Both models produce strong F1 scores of .88/.89 with minor differences in their trade-off with precision and recall. Another way to compare the effectiveness of their models is by using their cross-validation score (CV). Utilizing k-fold cross-validation, I divided the dataset into 5 subsets or folds and trained them using the folds against each other for validation. The purpose is to gather a more reliable estimate of the model's performance and stability, especially when different models produce similar results.

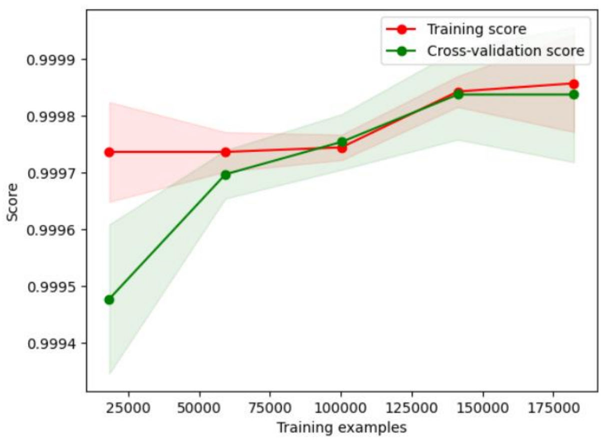

Figure 10.1 – RF-FFNN-LR CV score (Mean = .99983)

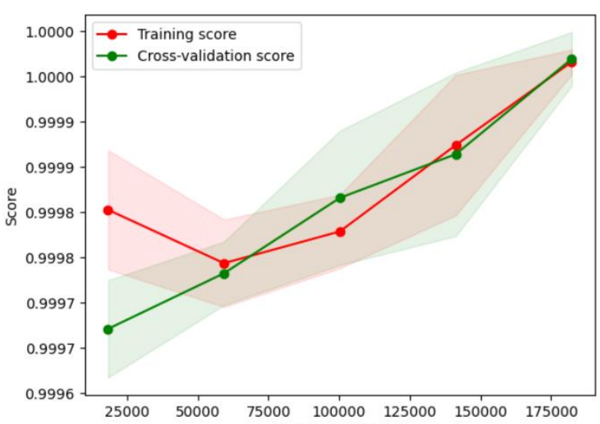

Figure 10.2 – RF-CNN-LR CV score (Mean = .99996)

Looking at both graphs, they show stability and high performance. The variability or standard deviation (shaded area) narrowing in the RF-CNN-LR model indicates consistency, unlike the prior model which also has a higher training score indicative of overfitting tendencies. Although the FFNN meta-model utilizes fewer resources, the CNN meta-model showcases marginal improvements in all metrics at the cost of a few minutes of training time.

Real-world Application

The models explored in this study have several real-world applications, specifically in anomaly detection. The metrics prove that these models can be tools for applications outside of credit card detection and can serve in other sectors like anomalies in medical billing or detecting breaches or unauthorized access attempts.

Moving Forward

The results of this study have reinforced the foundation of fraud detection models and insights into areas of improvement. The following strategies can be pursued to address the observed limitations:

- Model simplification –
  While performance in hybrid models like RF-CNN resulted in a strong balance in precision and recall, its complexity might propose new challenges in use. Simplifying architecture or focusing on other techniques could enhance performance.

- Additional resampling techniques –
  While under-sampling the majority class proved useful, techniques like SMOTE did not. Additionally, techniques could be experimented with techniques such as Adaptive Synthetic Sampling (ADASYN) to feed more meaningful data.

- Exploring other ensemble techniques –
  Leveraging other methods like bagging or boosting could utilize the strengths of models better. Techniques such as ADABoost, XGBoost, or LightGBM target gradient boosting.

CONCLUSION

The findings of this study highlight the effectiveness of hybrid and meta-models in identifying patterns with large datasets while maintaining efficient resource utilization. These models proved PCA’s ability to preserve privacy without compromising most of the evaluation process. By comparing these results with past studies, I was able to reaffirm the importance of refinement in fraud detection.

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